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INTRA-EQUILIBRIUM AND INTER-EQUILIBRIUM ANALYSIS IN CAPITAL MARKET THEORY: A CLARIFICATION

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1. INTRODUCTION

THE THEORY of capital asset pricing under uncertainty has received considerable attention in the literature on finance following the initial formulations of Sharpe (1964), Lintner (1965), and Mossin (1966). However, the interpretation of the capital asset pricing model in terms of its implications for allocation of resources in the economy, has been the subject of much controversy.

A particular case of the general question of resource allocation in the context of the capital asset pricing model is the problem of optimal investment and financing policies of the firm. In this context, the issues raised have been twofold: What is the effect of leverage on the market value of the firm? What should be the cutoff rate for investment decisions i.e., the cost of capital for a project?

Modigliani and Miller [MM] (1958) and (1963) raised these issues in a partial equilibrium setting in both a no-tax world and a world with taxes, assuming perfect capital markets. The question naturally arises as to the linkage between the Modigliani-Miller analysis and the more recent theory of capital asset pricing. Lintner (1965), Hamada (1969), Mossin [(1969) and (1973)] examine the Modigliani-Miller propositions in the context of the capital asset pricing model and claim that the MM propositions hold only as an approximation. They argue that MM fail to consider changes in the aggregate level of investment in the economy as a result of a firm's decisions and the resulting changes in the relevant parameters of the capital asset pricing model.¹

If these analyses were valid, they would be fairly damaging to much of the theory of finance. In particular, even under perfect competition, the concept of risk independence—that a project can be judged on its own merit, without considering its effect on the rest of the firm—would be violated.² Further, the separation property of firm level decision making rules will be destroyed and firms cannot evaluate a decision without considering its effect on existing investments.

Fortunately, these positions expressed by Lintner, Mossin, Hamada and Kumar, stem from an inappropriate use of equilibrium analysis. There is a fundamental distinction between intra-equilibrium analysis and inter-equilibrium analysis which,

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1. In a recent paper, Kumar (1974) attempts to show that the MM proposition III that the cost of capital is independent of financing and the level of investment in a no-tax world, is violated, when the changes in aggregate parameters are taken into account.

2. See Myers (1968) for a discussion and proofs of risk independence in a state-preference framework.

if taken into account, resolves the above contradictions. The aim of this paper is to demonstrate that by correct use of these types of analysis the MM analysis and the capital asset pricing model, are consistent with each other.

2. COMPETITIVE EQUILIBRIUM AND COMPARATIVE STATICS

We first briefly review the two types of equilibrium analysis before applying them to the specific case of capital asset pricing in a mean-variance world. An equilibrium is said to exist in an economy if all the excess demands for different commodities, derived from the preferred actions of the participants, are non-positive.^{3,4} Once equilibrium is reached, intra-equilibrium relations between the variables in the system can be specified. For example, in the capital asset pricing model, a relationship exists between risk and return; in the MM theory of capital structure the relationship between the values of two firms, in the same risk class but with different capital structures, is specified. The specification of these relationships is dependent on the existence of a competitive equilibrium. No such specification is possible in a situation out of equilibrium, or even in the approach to equilibrium. For example, the MM arbitrage proof on the irrelevance of capital structure starts from "initial conditions" not in equilibrium and guarantees the equality of values of the levered and unlevered firms only at equilibrium.

As opposed to the analysis at equilibrium, there is the concept of comparative statics, (i.e., inter-equilibrium analysis) a methodology for comparing the values of variables in alternative equilibria resulting from changes in parameter values in the system. For example, in the capital asset pricing model, a change in the number of individuals in the economy should be and is analyzed in a comparative statics setting.⁵

We can observe two phases in the development of capital market theory under uncertainty that are consistent with the two types of equilibrium analysis. In the first phase, assuming maximization behavior on the part of individuals and the existence of equilibrium, a set of equations is solved to provide cross-sectional relations in equilibrium under a set of specific assumptions. This relation is in exact quantitative form and can, therefore, be tested empirically. The second phase, deals with the effect of changes in parameters on the equilibrium values of the variables in the system. The previous set of assumptions is not changed except that now one of the previously given parameters changes in a particular direction. This phase provides us with a meaningful set of predicted changes resulting from parameter changes.⁶

The recent controversy results from a confusion of theorems derived within an equilibrium with those obtained by considering alternative equilibria through

3. See Arrow and Hahn (1971) p. 23 for a mathematical definition and discussion.

4. An equilibrium is defined to possess perfect stability if from any initial conditions all the variables in the system approach their equilibrium values in the limit as time becomes infinite. Arrow and Hahn (1971) pp. 271-272 provide a rigorous mathematical definition.

5. See Lintner (1970), Rubinstein (1973) for examples of this type of analysis.

6. In empirical testing, relationships within an equilibrium can be estimated using cross-sectional data, while time-series data must be employed to test comparative statics results. In general, the results of the cross-sectional theorems are not interchangeable with those of the time-varying ones.

comparative statics. The derivation of decision rules for corporate investment, from the general equilibrium in the capital market is the prime example of this misunderstanding.

In the intra-equilibrium setting firm investment decisions and individual portfolio decisions are undertaken simultaneously.⁷ Firms enter the capital market with various proposals for investment in different "technologies", the characteristics of which are specified in terms of their means, variances and covariances with other technologies. Individuals make their portfolio decisions regarding both investment and borrowing-lending simultaneously. The market portfolio is iteratively determined, by choosing investments that have greater present market values than the level of investment undertaken, at the particular market price of risk chosen. At equilibrium, the aggregate investment of individuals must equal the value of the market portfolio, and aggregate borrowing and lending net out to zero. In this equilibrium, the market price of risk and the riskless rate of interest are determined simultaneously, along with the levels of investment in different technologies by various firms. In equilibrium, every technology has to satisfy the pricing relationship of the capital asset pricing model. In rate of return terms, all technologies have to satisfy⁸

$$\bar{R}_j \geq R_f + \lambda \sigma_{mj} \quad (1)$$

where $\bar{R}_j$ is the expected return on the j th technology, R_f is the riskless rate of interest, λ represents the market price of risk in percentage terms and σ_{mj} is the measure of the risk of the j th technology. Equation (1) is the criterion for acceptance/rejection of investment in a particular technology, *at the margin*.

The alternative and incorrect approach is to apply an inter-equilibrium analysis to derivation of firm investment decision rules. This type of analysis considers the change in the values of parameters, as a result of additional investment by a particular firm in a given technology. In the movement from one equilibrium to another, the resulting change in the capital market parameters, the price of risk, the riskless rate of interest, and the return on the market portfolio as well as the measure of risk of the particular technology are considered. The "second-order" effects are included in the criterion for acceptance/rejection of additional investment.⁹

$$\bar{R}_j = R'_f + \lambda' \sigma'_{mj} \quad (2)$$

where R'_f , λ' and σ'_{mj} are the new riskless rate of interest, market price of risk and measure of risk respectively, following the additional investment by the firm. In general $R'_f \neq R_f$, $\lambda' \neq \lambda$, $\sigma'_{mj} \neq \sigma_{mj}$. The acceptance/rejection of a "project" by a firm depends on a) its effect on the market price of risk and the riskless rate of interest through second-order changes, and b) the nature of the returns from the particular

7. See Black (1972) for an articulation of this position.

8. See, for example, Sharpe (1964) and Lintner (1965).

9. Many of the proponents of this position neglect to consider the change in the riskless rate of interest following an increase in the level of investment in a particular technology. In principle, the riskless rate of interest must be considered on the same basis as the market price of risk.

firm's investment in other technologies through the covariance between the returns of the project and other investments undertaken by the firm. For an expansion in the level of investment in the (only) technology used by a particular firm, the variance of the technology's returns per unit investment, takes the place of the covariance term. This implies, for instance, that the cut-off point for investment in a given technology become more and more binding, as the level of investment in the technology is increased, by a particular firm.¹⁰

Some authors proposing the latter view contend that the change in the prevalent price of risk, λ , is a second order effect and is negligible. The effect of the returns from existing investments by the firm, is more difficult to explain away, since it may not be negligible in many cases.

As we briefly stated before, these contradictions stem from a misapplication of equilibrium analysis. The capital asset pricing model is a relationship between returns and risk within a given competitive equilibrium, i.e. given the preferences of individuals and their endowments, the final set of values (prices) prescribed by the model will be reached irrespective of the starting values for prices in the iterative tâtonnement process. Individual entities in the market, both firms and individuals can act *as if* the equilibrium relationship will prevail, in making their decisions. Specifically a firm's investment decision will be made, taking the equilibrium relationship as given, in formulating the cutoff criterion for acceptance/rejection. It would be inconsistent with the notion of a stable competitive equilibrium for a firm to undertake expansions in the level of investment in an existing technology—one that existed in the previous equilibrium. This would violate the notion of competitive equilibrium, since it would imply a positive excess demand with respect to the previous equilibrium. At best, the previous set of prices may be regarded as a particular tâtonnement in the approach to an equilibrium.¹¹

After an equilibrium has been reached it is possible, in principle, to move to a new equilibrium, when a new set of investment opportunities is being considered. While the process of adjustment to the new equilibrium cannot be specified, in general, the values of λ (and R_f) will be different in the new equilibrium. Using the method of comparative statics, one can answer a specific set of questions regarding the effect of exogenous shocks on the equilibrium. However, this framework is inappropriate and even misleading when applied to answer questions pertaining to relationships between the endogenous variables of the system which are of concern to the firm in making its investment decisions.

3. THE LINTNER, MOSSIN AND HAMADA POSITIONS

The above points can best be illustrated by reference to the views of proponents of the interpretation in equation (2). Specifically, we will examine the views of Lintner, Mossin and Hamada, who have explicit positions on this issue.

In his first article on equilibrium in the capital market, Lintner (1965) makes the following "simplifying" assumption: that "aggregate market value of all other

10. See Mossin (1973) pp. 130, 136–138.

11. In principle, it is difficult to imagine an increase in the level of investment in an existing technology, given that an equilibrium initially exists i.e. a firm desiring an increase in investment in the particular technology would have signified its intent before the initial equilibrium.

stocks...are invariant to the capital budgeting decision of the *i*th company".¹² By making use of this assumption, Lintner provides a rule for investment decisions, but hastens to add that this assumption "is a convenience which only involves ignoring (generally small) second order feedback effects..."¹³

Lintner (1972) is more explicit, regarding his position on this issue, in his discussion of another paper. He asserts that "all the returns and specifically *all the covariances between all of the investments held in the new equilibrium* under the enlarged investment opportunity set must be fully reflected in the market prices and parameters of that new equilibrium position. And these covariances necessarily include those between the new and old assets of the *i*th firm *as well as* those between its new investments and the assets of other firms... On neither welfare nor share price grounds can Mossin's equation be criticized for including these intra-firm covariances..."¹⁴ (emphasis in original). Lintner's position negates the concept of risk independence by making the investment criterion firm specific. More importantly, it violates the notion of price taking in capital market equilibrium under uncertainty, by allowing market-wide parameters to change as a result of investment decisions by individual firms. In our view, the application of the comparative statics methodology suggested by Lintner is inappropriate unless the firm is assumed to have some positive monopoly power in the capital market, a position inconsistent with the spirit of the capital asset pricing model.

Another scholar who holds a similar position is Mossin. Though his initial description of market price setting was consistent with received theory (Mossin (1969)), he later modified his view and shares the Lintner position regarding decisions in an equilibrium setting. In relating the decision rules for investment by firms to the general equilibrium relationship of capital asset pricing theory, Mossin (1966, 1969, 1972, 1973) argues that the Modigliani-Miller theory of corporate finance, according to which proportional increases in the level of investment in a particular "risk class" lead to proportional increases in the market value of the firm, holds only as an approximation.¹⁵ He argues that by making an investment in a project, an individual firm affects both the market price of risk as well as the variance-covariance structure of returns in the market. In particular, the risk of the firm's investment increases more than proportionately (through the own variance term), while the return increases proportionately. Thus, even ignoring changes in the market price of risk, the value of the firm's increase in a technology increases less than proportionately in the Mossin analysis. Further, Mossin argues that through the changes in the variance-covariance structure of returns, the values of all other firms change, depending on their covariance with the returns from the investment in question, even if the market price of risk is assumed constant.

Mossin traces through the second order effects of investment in a technology by a particular firm and finds that the value of investment in a particular project

12. Lintner (1965), p. 698.

13. *ibid.*, p. 698.

14. Lintner (1972), p. 455.

15. According to the Modigliani-Miller theory of corporate investment, the cutoff point for all investments in a project in a particular risk class is constant across firms. An increase in any firm's level of investment leads to a proportionate change in the market value.

depends on which firm undertakes it. For example, if the project under consideration is positively correlated with existing projects, the larger the firm the stricter the cut-off criterion for acceptance or rejection; in equation (2), for example, σ_{mj} is greater for an existing firm than a new firm. Mossin regards the "decreasing returns to scale" in valuation with respect to investment by an individual firm as a blessing since otherwise "no finite equilibrium investment level exists".¹⁶ He concludes that a proper description of firm behavior would be as price takers with respect to the market price of risk, but realizes that the "proportionality rule" does not hold; "the proportionality assumption of the MM investment theory can be considered as an assumption of a rather crude sort of price taking behavior, while a more sophisticated firm would recognize the non-proportional change in [the measure of risk], but still act as a price taker with respect to [the market price of risk]. In many ways, the latter conception of the firm seems the more attractive."¹⁷

We argue that the Mossin characterization stems from a misapplication of the notion of equilibrium in a competitive setting. In a competitive equilibrium, it is not necessary that the equilibrium level of investment by *an individual firm* be uniquely determined. For example, it is sufficient that the *aggregate level* of investment by all firms be finite. In fact, in the theory of perfect competition under certainty, the level of investment and output by individual firms is indeterminate (between zero and the aggregate level), while the aggregate levels are given. Our earlier objections to the Lintner position regarding the misleading use of the notion of equilibrium apply to Mossin as well.

Hamada (1968) attempts to analyze the financing and investment decisions of the firm, using equilibrium concepts. He attempts to prove the Modigliani-Miller propositions regarding capital structure using capital market theory. However, he misinterprets the equilibrium conditions when he states that a "comparative statics framework will be used throughout this paper". The Modigliani-Miller propositions are stated under equilibrium conditions where the values of levered and unlevered firms are compared and the implications for the firms cut-off criterion are studied. The MM propositions are *not* stated in a comparative statics setting. If they were, the change in the riskless rate as a result of an increase in the use of debt will have to be considered, in the move from an equilibrium where a particular firm employs only equity to one where it levers up, for example. As a result, the value of the firm in the two alternative equilibria could, in principle, be different. But this is not the intent of the MM analysis. They compare the value of a levered and an unlevered firm *within* the same equilibrium. It is obvious therefore that Hamada's use of comparative statics analysis, to study MM's results stated within an equilibrium, is inappropriate.

Fortunately, Hamada's conclusions are right though for the wrong reason, since he assumes that the effect on the price of risk is negligible, which in principle *does* change.¹⁸ In his derivation of the cost of capital, Hamada states explicitly that his

16. Mossin (1973), p. 130.

17. *ibid.*, p. 130.

18. In a comment on Hamada, Kumar (1974) *does* take into consideration such effects, and arrives at questionable conclusions. In particular he claims that Modigliani-Miller's Proposition III that the cost of capital of a project is independent of the particular firm undertaking it, is inconsistent with the capital asset pricing model.

derivation stands only as an approximation in which the "secondary" effects are assumed small. He does not have to make that apology. His results hold exactly when interpreted in a "within equilibrium," rather than comparative statics, setting.

4. CONCLUSION

This paper emphasizes the distinction between two types of equilibrium analysis: the derivation of relationships between the variables at equilibrium (intra-equilibrium analysis) versus the change in the relationships in the new equilibrium arising from a change in one of the underlying parameters of the system (inter-equilibrium analysis). Since the capital asset pricing model is an example of the former type of analysis in a competitive capital market, the inappropriateness of using comparative statics methodology in such a setting are pointed out. When these elements are taken into account there is no inconsistency between capital asset pricing theory and the Modigliani-Miller propositions.

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